DS	**Session**	Stack	**Question Information**	Level 1 Challenge 50

Question description

Hassan gets a job in a software company in Hyderabad. The training period for the first three months is 20000 salary. Then incremented to 25000 salaries.

Training is great but they will give you a programming task every day in three months. Hassan must finish it in the allotted time. His teammate Jocelyn gives him a task to complete the concept of Postfix to Prefix Conversion for a given expression. can you help him?

Functional Description:

- Read the Prefix expression in reverse order (from right to left)
- If the symbol is an operand, then push it onto the Stack
- If the symbol is an operator, then pop two operands from the Stack
  Create a string by concatenating the two operands and the operator after them.
  string = operand1 + operand2 + operator
  And push the resultant string back to Stack
- Repeat the above steps until end of Prefix expression.

Constraints

the input should be a expressions

Input Format

Single line represents the postfixed expressions

Output Format

Single line represents the prefix expression

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
ABC/-AK/L-*
```

EXPECTED OUTPUT

```
*-A/BC-/AKL
```

Test Case 2

INPUT (STDIN)

```
AB+CD-*
```

EXPECTED OUTPUT

```
*+AB-CD
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
string postToPre(string
post_exp)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
5
```

Test Case 2

TOKEN COUNT

```
220
```

Test Case 3

NLOC

```
46
```

Code

solution.c

```c
#include <stdio.h>
#include <string.h>
#include <ctype.h>

char stack[100][1000];
int top = -1;

void push(char *s) { strcpy(stack[++top], s); }
char* pop() { return stack[top--]; }

int isOperator(char c) {
    return (c == '+' || c == '-' || c == '*' || c == '/' || c == '^');
}

int main() {
    char post_exp[1000];
    if (scanf("%s", post_exp) != 1) return 0;
    int len = strlen(post_exp);
    for (int i = 0; i < len; i++) {
        char ch = post_exp[i];
        if (isalnum(ch)) {
            char op[2] = {ch, '\0'};
            push(op);
        } else if (isOperator(ch)) {
            char op2[1000], op1[1000], res[2000];
            strcpy(op2, pop());
            strcpy(op1, pop());
            snprintf(res, sizeof(res), "%c%s%s", ch, op1, op2);
            push(res);
        }
    }
    printf("%s\n", pop());
    return 0;
}
```

Output

```
Problems  Output  Debug Console  Terminal  Ports        powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
AB+CD-*
*+AB-CD
PS C:\C Programming>
```